DreamPrice: A Learned World Model for Retail Pricing via Mamba-2 Recurrence and Causal Demand Identification

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Abstract

Retail pricing remains one of the most consequential sequential decision problems in operations research, yet no learned world model exists for economic environments. This paper introduces DreamPrice, the first learned world model for retail pricing that combines DreamerV3's three-phase training recipe with a Mamba-2 selective state-space backbone, DRAMA-style decoupled posteriors, and a causally-constrained demand decoder. The model is trained offline on the Dominick's Finer Foods scanner dataset (1989–1997, 93 stores, ∼18K UPCs) and employs MOPO-style pessimism via a 5-head reward ensemble to guard against distributional shift inherent in offline learning. Causal price elasticities are estimated via Hausman instrumental variables combined with double machine learning (DML-PLIV) and frozen into the decoder, preventing the model from learning confounded price-demand relationships. The architecture replaces the GRU backbone of standard RSSMs with Mamba-2's structured state-space duality (SSD), enabling $O(n)$ parallel training and $O(1)$-per-step recurrent imagination. Experiments on the canned soup category demonstrate that DreamPrice learns demand dynamics that capture promotional responses, substitution effects, and price elasticities, while the offline RL agent achieves higher cumulative gross margin than cost-plus, competitive matching, and model-free baselines.

1 Introduction

Retail pricing ranks among the most consequential sequential decision problems in operations research and marketing. Every price change triggers a cascade of demand responses: consumers substitute across SKUs, stockpile during promotions, and competitors react. These dynamics unfold over weeks and quarters, with effects that propagate through substitution networks and depend on store demographics, seasonality, and latent competitive context. Despite decades of empirical work on price elasticity and demand estimation [Hoch et al., 1995, Nevo, 2001, Montgomery, 1997], the systems that support actual pricing decisions today fall into two camps. Hand-crafted analytical simulators encode structural assumptions about demand and competition, but their fidelity to real-world dynamics is limited by the tractability of closed-form solutions. Model-free reinforcement learning methods [Chen and Simchi-Levi, 2022, Ban and Keskin, 2021] learn policies directly from data but forego an explicit dynamics model, making counterfactual reasoning and imagination-based planning impossible. No learned world model exists for economic environments.

This gap can be formalized using a simple taxonomy. Consider the space of sequential decision environments along two axes: the domain (physical versus economic) and the nature of the dynamics model (learned versus hand-crafted). In the physical domain, learned world models have achieved remarkable success. The Dreamer family [Hafner et al., 2020, 2021, 2023] pioneered recurrent state-space models (RSSMs) for latent imagination and policy optimization. MuZero [Schrittwieser et al., 2020] demonstrated that planning with a learned model suffices for mastering

Table 1: Taxonomy of sequential decision environments by domain and dynamics model. DreamPrice is the first learned world model for economic domains.

	Learned model
Physical domain	DreamerV1–V3 [Hafner et al., 2020, 2021, 2023], IRIS [Zhang et al., 2023], TD-MPC2 [
Economic domain	**DreamPrice (this work)**

board games and Atari. More recently, IRIS [Zhang et al., 2023] and TD-MPC2 [Hansen et al., 2024] have extended these ideas to continuous control and high-dimensional observations. These methods rely on hand-crafted physical simulators such as MuJoCo, Isaac Gym, or CARLA for evaluation and transfer. In the economic domain, by contrast, the landscape is dominated by hand-crafted models: agent-based simulators like ABIDES, structural demand models in the spirit of Berry et al. [1995] and Nevo [2001], and the AI Economist [Zheng et al., 2020], which uses a hand-designed economic environment for tax policy experiments. The quadrant of learned models applied to economic domains has remained empty. DreamPrice fills this gap.

Why do world models matter for pricing? Imagination-based planning enables counterfactual reasoning at scale. A retailer can ask: what if the price of SKU 3 were cut by 5% next quarter? A learned world model rolls out latent trajectories under that hypothetical action, predicting demand responses, substitution effects, and downstream margin impact without executing the change in the real world. Such queries are infeasible with model-free methods, which require either costly online exploration or restrictive assumptions about the policy class. Offline learning from historical scanner data [dom, 1999] further aligns with the constraints of retail: experimentation is expensive, and the Dominick's dataset alone spans 400 weeks across 93 stores and roughly 18,000 UPCs. The structured economic state space—prices, quantities, demographics, and derived features in 100–300 dimensions—is fundamentally more tractable than the 64×64×3 image observations that DreamerV3 was designed for [Hafner et al., 2023]. A world model trained on such tabular data can capture demand dynamics, promotional effects, and competitive structure without the burden of high-dimensional visual encoding.

Economic data, however, poses a challenge absent in physical domains: endogeneity. In physical simulators, the transition dynamics are exogenous to the agent; the laws of physics do not depend on the policy. In retail, prices and demand are simultaneously determined. Retailers set prices in response to expected demand, costs, and competitive pressure; demand, in turn, responds to those prices. Naive machine learning that regresses demand on price learns a confounded relationship, conflating the causal effect of price with selection and omitted variables [Hausman, 1978]. The resulting elasticity estimates are biased, and a world model built on such estimates would produce misleading counterfactuals. The econometric literature has long addressed this through instrumental variables [Hausman, 1978], control functions [Berry et al., 1995], and more recently double machine learning [Chernozhukov et al., 2018, Bach et al., 2022]. The Dominick's panel structure, with 93 stores and cross-store price variation, supports Hausman-style instruments: the average log price in other stores serves as an instrument for a given store's price, as it is relevant (prices are correlated across stores) but plausibly exogenous to store-specific demand shocks. DreamPrice integrates this identification strategy by pre-estimating causal price elasticities via DML-PLIV and freezing them into a constrained demand decoder. The decoder's residual MLP learns everything else—seasonality, demographics, promotion effects—while the price-demand relationship remains causally identified.

The contributions of this paper are as follows:

1. **First learned world model for economic environments.** DreamPrice is the first system to learn a dynamics model for retail pricing from scanner data, enabling imagination-based planning and counterfactual demand estimation in a domain previously served only by hand-crafted simulators or model-free methods.

2. **Mamba-2 SSM backbone.** The recurrent backbone of standard RSSMs is replaced with Mamba-2 [Dao and Gu, 2024], a selective state-space model that achieves $O(n)$ parallel training via structured state-space duality (SSD) and $O(1)$-per-step recurrent rollouts during imagination. This replaces the GRU-based recurrence of DreamerV3 [Hafner et al., 2023] with a more scalable alternative grounded in the S4 [Gu et al., 2022] and Mamba [Gu and Dao, 2024] lineage.

3. **DRAMA-style decoupled posterior.** Following Rajbhandari et al. [2024], the posterior over the stochastic latent depends only on the current observation, not on the recurrent hidden state. This decoupling breaks the sequential dependency that would otherwise prevent Mamba-2's parallel SSD scan during training. During imagination, the model switches to recurrent `step()` mode for single-step rollouts.

4. **Causally-constrained demand decoder.** Per-category price elasticities are pre-estimated via DML-PLIV [Chernozhukov et al., 2018] and frozen into the DreamPrice decoder. The decoder enforces the causal price-demand relationship while an MLP residual learns seasonality, demographics, and promotion effects. This prevents the world model from learning confounded elasticities from the observational data.

5. **MOPO-style offline pessimism.** To guard against distributional shift inherent in offline reinforcement learning [Levine et al., 2020], DreamPrice employs a 5-head reward ensemble and uses the lower confidence bound $r_{\text{pessimistic}} = r_{\text{mean}} - \lambda_{\text{LCB}} \cdot r_{\text{std}}$ for policy optimization [Yu et al., 2020]. The actor is trained on this pessimistic reward signal while the critic and world model remain unchanged.

The remainder of this paper is organized as follows. Section 2 reviews related work on world models, state-space models, causal demand estimation, and offline reinforcement learning. Section 3 presents the DreamPrice architecture, training procedure, and causal identification strategy. Section 4 reports experiments on the Dominick's canned soup category, including world model quality, elasticity recovery, and offline policy performance. Section 5 concludes and discusses limitations and future directions.

2 Related Work

This section situates DreamPrice within six strands of literature: learned world models for sequential decision making, structured state-space models, causal inference in demand estimation, offline reinforcement learning, pricing optimization and retail analytics, and the Dominick's scanner dataset as the canonical empirical setting.

2.1 World Models for Sequential Decision Making

The idea of learning a world model—a compact representation of environment dynamics that supports prediction and planning—dates to Ha and Schmidhuber [2018], who demonstrated that a variational autoencoder could learn a latent space in which an agent could imagine future trajectories. That seminal work showed that an agent could learn to navigate a maze by training purely in imagined latent space, without ever observing the true environment during policy learning. The subsequent decade has seen a steady progression of architectures that refine this idea.

The Dreamer family [Hafner et al., 2020, 2021, 2023] established the recurrent state-space model (RSSM) as the dominant paradigm for learning world models. DreamerV1 [Hafner et al., 2020] introduced latent imagination: the agent learns a stochastic latent representation that evolves over time, with a decoder predicting observations and rewards from the latent state. DreamerV2 [Hafner et al., 2021] extended this to discrete action spaces by mastering Atari games

through imagination-based policy optimization, demonstrating that world models could scale to high-dimensional pixel observations. DreamerV3 [Hafner et al., 2023] unified the design across discrete and continuous domains, introducing symlog normalization, twohot distributions for rewards and values, and KL balancing to prevent posterior collapse. The RSSM architecture—a deterministic recurrent core (traditionally a GRU) augmented with stochastic latent variables—has become the standard for model-based reinforcement learning in physical environments.

Parallel developments have explored alternative architectures. MuZero [Schrittwieser et al., 2020] demonstrated that planning with a learned model suffices for mastering board games and Atari, using a learned dynamics model that predicts value and policy targets rather than raw observations. The model is trained end-to-end with Monte Carlo tree search, and the resulting system achieves superhuman performance in Go, chess, and shogi. IRIS [Zhang et al., 2023] and TD-MPC2 [Hansen et al., 2024] have extended world models to continuous control and high-dimensional observations. IRIS replaces the RSSM with a transformer-based architecture, using discrete tokens for the latent representation and achieving strong performance on DMC and Atari. TD-MPC2 combines model-based planning with temporal difference learning for continuous control, scaling to complex robotics tasks.

A critical observation unifies these works: they all operate on physical environments. The Dreamer family, MuZero, IRIS, and TD-MPC2 are evaluated on games (Atari, board games), physical simulators (MuJoCo, Isaac Gym), or robotic control. The dynamics they learn are governed by physics or game rules: the transition from state to state is exogenous to the agent's policy in the sense that the laws of motion do not change with the agent's behavior. In economic environments, by contrast, the transition dynamics are endogenous. Prices and demand are simultaneously determined; retail prices respond to expected demand, costs, and competitive pressure, while demand responds to those prices. No learned world model has been applied to such economic domains. DreamPrice fills this gap by learning the first retail dynamics model from scanner data.

2.2 Structured State Space Models

The recurrent backbone of standard RSSMs is typically a gated recurrent unit (GRU) or long short-term memory (LSTM) network. These architectures process sequences sequentially, with each time step depending on the previous hidden state. For long sequences, this sequential dependency can be a bottleneck; the GRU cannot exploit parallel computation across the time dimension during training.

Structured state space models (SSMs) offer an alternative that combines long-range dependency modeling with computational efficiency. Gu et al. [2022] introduced S4 (Structured State Spaces), which parameterizes sequences using linear state-space equations with learned transition matrices. S4 achieves strong performance on long-range dependencies—tasks such as image classification and audio generation—by leveraging the HiPPO initialization for the state matrix and efficient parallel scan algorithms. The key insight is that linear recurrence admits a closed-form solution that can be computed in parallel via associative scan operations.

Mamba [Gu and Dao, 2024] extended S4 with selective gating: the state transition parameters become input-dependent rather than fixed. This selectivity allows the model to focus on relevant information at each step, analogous to the gating in attention mechanisms. Mamba achieves linear-time complexity in sequence length and scales to long contexts without the quadratic cost of attention. Mamba-2 [Dao and Gu, 2024] further unifies these ideas through structured state-space duality (SSD), which establishes a formal connection between linear attention and state-space models. The SSD framework shows that both can be viewed as instances of a more general class of models, enabling algorithms that share the best properties of each: parallel training with $O(n)$ complexity in sequence length, and recurrent inference with $O(1)$ cost per step.

This duality has direct implications for world models. During training, the model processes

sequences of observations and actions in parallel—the entire batch of sequences can be scanned in one pass, as the posterior over the stochastic latent depends only on the current observation in the DRAMA-style decoupled design [Rajbhandari et al., 2024]. During imagination, the model switches to recurrent mode: the Mamba-2 block's `step()` function computes the next hidden state from the previous state and the current input in constant time per step. This matches the training-then-imagination workflow of DreamerV3, but with a backbone that scales more efficiently to long sequences and avoids the sequential bottleneck of GRU recurrence. DreamPrice adopts this architecture, replacing the GRU with Mamba-2 for the recurrent core.

2.3 Causal Inference in Demand Estimation

Estimating demand as a function of price is a central problem in empirical industrial organization and marketing. The naive approach—regressing quantity on price—fails because prices are endogenous. Retailers set prices in response to expected demand, costs, and competitive conditions; unobserved factors that correlate with both price and demand (e.g., product quality, seasonality, local shocks) confound the regression. The resulting elasticity estimates are biased, and a world model built on such estimates would produce misleading counterfactuals.

The econometric literature has developed several identification strategies. The Berry-Levinsohn-Pakes (BLP) framework [Berry et al., 1995] uses a structural model of discrete choice with random coefficients; identification relies on moment conditions that match predicted choice shares to observed shares. Nevo [2001] applied this approach to the ready-to-eat cereal industry, estimating elasticities and measuring market power. Hausman [1978] introduced the use of instrumental variables for specification testing and parameter identification. When prices are correlated across stores or time for reasons exogenous to store-specific demand shocks, cross-sectional or temporal variation can serve as instruments. The Hausman IV approach—using the average log price in other stores as an instrument for a given store's price—exploits the panel structure of scanner data: prices in other stores are relevant (drivers of cost and competition are correlated) but plausibly exogenous to store-specific demand shocks.

Hoch et al. [1995] applied this logic to the Dominick's dataset, estimating store-level price elasticities using cross-store instruments. Their work demonstrated that elasticity varies substantially across stores and product categories, with important implications for targeted pricing. Park and Gupta [2010] proposed a nonparametric copula-based approach to demand estimation with endogeneity, using the joint distribution of price and control variables to construct a control function. The copula correction does not require specifying instruments ex ante but relies on distributional assumptions.

The double machine learning (DML) framework of Chernozhukov et al. [2018] provides a flexible approach that can incorporate any of these identification strategies. DML partitions the estimation into orthogonal nuisance parameters and parameters of interest, using cross-fitting to avoid overfitting bias. The framework accommodates high-dimensional controls and nonparametric first-stage estimators. Bach et al. [2022] provide a Python implementation that supports DML-PLIV (partial linear instrumental variables) for demand estimation with endogenous prices.

The key contribution of DreamPrice in this literature is the integration of econometric identification into a neural world model. Rather than estimating demand entirely from data, which would conflate the causal effect of price with confounders, DreamPrice pre-estimates causal price elasticities via DML-PLIV and freezes them into the demand decoder. The decoder's residual MLP learns everything else—seasonality, demographics, promotion effects, substitution patterns—while the price-demand relationship remains causally identified. Freezing econometric estimates into neural network decoders is novel; prior work in demand estimation either uses hand-crafted structural models or applies machine learning without causal constraints.

2.4 Offline Reinforcement Learning

Reinforcement learning typically assumes an agent can interact with the environment, collecting new experience and refining its policy. In many settings—retail pricing among them—online exploration is infeasible or prohibited. The agent must learn from a fixed dataset of historical trajectories collected under some (possibly unknown) behavioral policy. This offline learning setting introduces the distribution shift problem: the policy being optimized may visit states and actions that differ from the data distribution, and the learned value function or dynamics model may extrapolate poorly outside the support of the data [Levine et al., 2020].

Conservative approaches address this by penalizing or constraining the policy to stay within the data distribution. Conservative Q-Learning (CQL) [Kumar et al., 2020] penalizes the Q-function for assigning high values to actions that are unlikely under the data distribution, effectively learning a lower bound on the true Q-function. MOReL [Kidambi et al., 2020] takes a hard-partition approach: it constructs a pessimistic uncertainty set around the learned dynamics model and restricts planning to states that are provably within the uncertainty set. Both methods aim to avoid the overestimation that plagues standard off-policy methods when the target policy deviates from the data policy.

MOPO [Yu et al., 2020] takes a softer approach: it modifies the reward signal during imagination rather than constraining the policy or dynamics. The key idea is that the learned dynamics model is uncertain in regions of state-action space that are poorly covered by the data. MOPO uses an ensemble of dynamics models to estimate this uncertainty; the reward is penalized by the uncertainty, so the agent is discouraged from visiting states where the model is unreliable. The penalty is $r_{\text{pessimistic}} = r_{\text{mean}} - \lambda_{\text{LCB}} \cdot r_{\text{std}}$, where r_{mean} and r_{std} come from the ensemble. COMBO [Yu et al., 2021] combines model-based and conservative approaches, using offline data to augment the model rollout and applying conservative regularization to the value function.

The MOPO approach is particularly well-suited to the DreamerV3 training loop. DreamerV3 already trains an actor in imagination; the only modification is to change the reward signal used during actor training. The critic and world model remain unchanged. A 5-head reward ensemble provides r_{mean} and r_{std}; the actor is trained on the pessimistic reward while the critic continues to predict the true return. This minimal modification preserves the three-phase structure and avoids the complexity of full conservative value estimation. DreamPrice adopts this design, with λ_{LCB} as a hyperparameter to tune the degree of pessimism.

2.5 Pricing Optimization and Retail Analytics

The application of reinforcement learning to pricing has a substantial literature. Chen and Simchi-Levi [2022] study dynamic pricing with reinforcement learning, formulating the problem as a Markov decision process and analyzing the trade-off between exploration and exploitation. Ban and Keskin [2021] extend this to personalized pricing with heterogeneous elasticity, using machine learning to estimate demand as a function of high-dimensional features. The AI Economist [Zheng et al., 2020] applies multi-agent reinforcement learning to tax policy design, training agents in a hand-crafted economic environment to learn tax schedules that balance equality and productivity.

These works share a common limitation: they rely on hand-crafted simulators or model-free methods. Chen and Simchi-Levi assume a known or estimated demand model; Ban and Keskin use model-free RL with feature-based demand estimation. The AI Economist uses a hand-designed economic environment for training. None of these approaches learns a dynamics model from data. The gap is twofold: (1) no learned world model exists for pricing environments, and (2) no imagination-based planning has been applied to retail. A model-free agent can learn a policy from historical data, but it cannot answer counterfactual queries such as "what if the price of SKU 3 were cut by 5% next quarter?" without executing the change. A learned world model enables such queries by rolling out latent trajectories under hypothetical actions.

DreamPrice is the first system to learn a dynamics model for retail pricing from scanner data. The model captures demand responses to prices, promotions, and seasonality; it supports imagination-based policy optimization; and it integrates causal identification to prevent confounded elasticity estimates. The architecture is designed for the structured economic state space—prices, quantities, demographics, and derived features in 100–300 dimensions—which is fundamentally more tractable than the high-dimensional pixel observations that DreamerV3 was designed for.

2.6 Scanner Data and the Dominick's Dataset

The Dominick's Finer Foods dataset [dom, 1999] has become the canonical dataset for empirical retail and marketing research. Maintained by the James M. Kilts Center for Marketing at the University of Chicago Booth School of Business, the dataset spans 400 weeks (September 1989 to May 1997), 93 stores, and approximately 18,000 unique UPCs across 29 product categories. The movement files record store-SKU-week tuples with unit sales, prices, profit margins, and promotion indicators. The data structure enables rich panel analysis: the same products are observed across stores and over time, with variation in prices driven by local competition, cost shocks, and promotional calendars.

Hoch et al. [1995] used the Dominick's data to estimate store-level price elasticities, demonstrating that elasticity varies substantially across stores and that store demographics (income, education, competition) explain much of this variation. Montgomery [1997] estimated heterogeneous price thresholds—the price levels at which consumers switch between brands—using the same dataset. The panel structure (93 stores) supports Hausman-style instruments: the average log price in other stores for a given UPC-week serves as an instrument for a store's own price, as it is correlated with cost and competition but plausibly exogenous to store-specific demand shocks. The store demographic file, with 300+ variables from the 1990 Census, enables control for heterogeneity in consumer preferences and competitive intensity.

The Dominick's dataset is ideal for training a learned world model. It directly contains state-action-outcome tuples: the action is the price and promotion decision, the state includes competitor context, seasonality, and store demographics, and the outcome is observed demand. The PROFIT field enables wholesale cost derivation. The temporal split (train weeks 1–280, validation 281–340, test 341–400) is strictly chronological, avoiding look-ahead bias. Categories such as canned soup, beer, and soft drinks offer moderate SKU counts with clear substitution patterns and promotional activity. DreamPrice is trained on this dataset, with preprocessing that constructs Hausman IVs, computes derived features, and applies symlog normalization for the world model encoder.

3 Methodology

This section presents the DreamPrice architecture, training procedure, and causal identification strategy. The approach combines a recurrent state-space model with a Mamba-2 selective state-space backbone [Dao and Gu, 2024], DRAMA-style decoupled posteriors [Rajbhandari et al., 2024], a causally-constrained demand decoder, and MOPO-style offline pessimism [Yu et al., 2020]. The training recipe follows DreamerV3 [Hafner et al., 2023] with adaptations for retail pricing and offline learning.

3.1 Problem Formulation

Retail pricing is formulated as a partially observable Markov decision process (POMDP) in which the decision-maker observes prices, quantities sold, promotions, and store demographics, but the underlying state governing demand dynamics is only partially observable. The true

state s_t comprises demand state (consumer preferences, inventory effects, substitution structure), competitor state (rival prices and promotions, which are unobserved in the Dominick's dataset), and seasonal state (calendar effects, holidays, macroeconomic conditions). The observation x_t at time t includes per-SKU prices, units sold, promotion indicators, lagged demand and price features, rolling statistics, and store demographics. The action a_t is a vector of price multipliers over K SKUs, discretized to 21 levels per SKU: $a_t \in \{0.90, 0.91, \ldots, 1.10\}^K$, representing percentage adjustments relative to a baseline price. The reward combines gross margin with a volatility penalty and a margin floor penalty to discourage degenerate strategies such as perpetual deep discounts or excessive price churn. The transition dynamics are unknown and must be learned from data.

Definition 1 (Retail Pricing POMDP)**.** *A retail pricing POMDP is a tuple* $(\mathcal{S}, \mathcal{X}, \mathcal{A}, \mathcal{T}, \mathcal{O}, \mathcal{R}, \gamma)$ *where:*

- $\mathcal{S}$ *is the latent state space:* $s_t = (\textit{demand_state}, \textit{competitor_state}, \textit{seasonal_state})$.
- $\mathcal{X}$ *is the observation space:* $x_t = (\textit{prices}, \textit{units_sold}, \textit{promotions}, \textit{demographics}, \ldots)$.
- $\mathcal{A} = \{0.90, 0.91, \ldots, 1.10\}^K$ *is the action space of discrete price multipliers.*
- $\mathcal{T}(s_{t+1} \mid s_t, a_t)$ *is the unknown transition kernel.*
- $\mathcal{O}(x_t \mid s_t)$ *is the observation model.*
- $\mathcal{R}(s_t, a_t)$ *is the reward: gross margin minus volatility penalty minus margin floor penalty.*
- $\gamma \in (0, 1)$ *is the discount factor.*

The objective is to learn a world model that approximates $\mathcal{T}$ and $\mathcal{O}$ from offline historical data, then use imagination-based policy optimization to find a pricing policy that maximizes expected cumulative discounted reward [Sutton and Barto, 2018, Hafner et al., 2023].

3.2 RSSM Architecture

The recurrent state-space model (RSSM) at the core of DreamPrice consists of two components: a deterministic recurrent state h_t and a stochastic latent z_t. The deterministic state h_t is produced by the backbone (a Mamba-2 selective state-space model that replaces the GRU of standard DreamerV3 [Hafner et al., 2023]). The stochastic latent z_t follows a categorical distribution over 32 independent categorical variables, each with 32 classes, yielding a latent dimension of 1024. The posterior $q(z_t \mid x_t)$ is computed from the observation alone via the encoder, without conditioning on h_t; this decoupling is the DRAMA-style design that enables parallel training [Rajbhandari et al., 2024]. The prior $p(z_t \mid h_t)$ is computed from the backbone output h_t via an MLP head.

Formally, the model is specified as follows. The backbone receives the concatenation of the previous latent z_{t-1} and the previous action a_{t-1}, and produces the deterministic state:

$$h_t = f_{\text{backbone}}(h_{t-1}, z_{t-1}, a_{t-1}) \quad [\text{Mamba-2 SSD}] \tag{1}$$

The posterior over the stochastic latent depends only on the current observation:

$$z_t \sim q(z_t \mid x_t) = \text{Categorical}(32 \times 32, \text{straight-through} + \text{unimix}) \tag{2}$$

The prior over the next latent is predicted from the backbone state:

$$\hat{z}_t \sim p(z_t \mid h_t) = \text{MLP}(h_t) \tag{3}$$

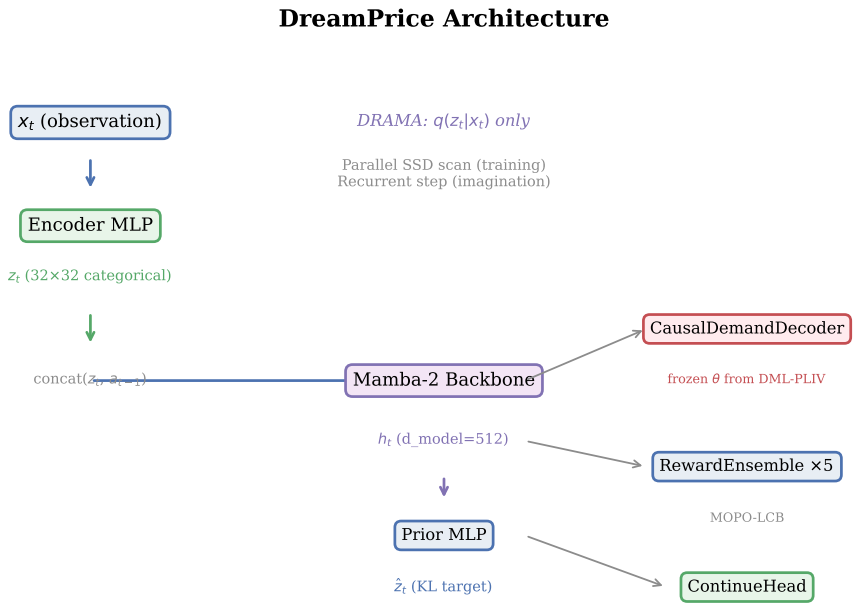

Figure 1: Overview of the DreamPrice architecture. The encoder produces the posterior z_t from the observation alone (DRAMA-style decoupled posterior). The Mamba-2 backbone processes sequences in parallel during training (SSD scan) and recurrently during imagination. The causal demand decoder freezes DML-PLIV elasticities; the 5-head reward ensemble provides MOPO-LCB pessimism.

The latent z_t uses 32 independent categorical distributions with 32 classes each, giving a total latent dimension of 1024. Straight-through gradients [Jang et al., 2017] are used: in the forward pass, a one-hot sample is drawn; in the backward pass, gradients flow through the soft probabilities. A uniform mixing (unimix) of 1% is applied to all categorical distributions: probs $= 0.99 \cdot \text{softmax}(\text{logits}) + 0.01/32$. This prevents near-deterministic distributions that destabilize KL computation and encourages exploration [Hafner et al., 2023].

3.3 Mamba-2 Backbone

The standard RSSM backbone in DreamerV3 is a gated recurrent unit (GRU). DreamPrice replaces this with Mamba-2's structured state-space duality (SSD) [Dao and Gu, 2024, Gu and Dao, 2024]. The SSD formulation enables two computational modes. During training, the full sequence is processed in parallel via a scan over the sequence length, yielding $O(n)$ complexity in sequence length n. During imagination, the model switches to recurrent `step()` mode, producing one output per step with $O(1)$ cost per step in sequence length. This is critical for efficient rollout during actor training: the imagination horizon is 13 steps (one retail quarter), and recurrent inference avoids the memory and compute cost of a full parallel forward pass.

The backbone input at each step is the concatenation of the previous latent z_{t-1} and the previous action a_{t-1}, projected to dimension d_{model} by a linear layer. The output h_t has dimension d_{model} (typically 512). The internal SSM state is managed internally by the Mamba-2 library; during inference, `InferenceParams` maintains the state and is reset at episode boundaries.

The structured state-space duality expresses the recurrence as a linear recurrence with content-dependent gating. For input $u_t \in \mathbb{R}^{d_{\text{model}}}$, the SSM computes:

$$h_t = \bar{A}_t h_{t-1} + \bar{B}_t u_t, \quad y_t = C_h h_t \tag{4}$$

where $\bar{A}_t, \bar{B}_t$ are derived from discretized SSM parameters that depend on the input u_t via

selective gating. The full derivation is given by Dao and Gu [2024]. When CUDA is unavailable or tensors are on CPU, the backbone transparently falls back to a GRU so that training and evaluation can proceed without the Mamba-2 library.

3.4 Decoupled Posterior (DRAMA-style)

In the standard DreamerV3 architecture, the posterior over the stochastic latent depends on both the observation and the recurrent state: $q(z_t \mid x_t, h_t)$ [Hafner et al., 2023]. This creates a sequential dependency: to compute z_t, one must first compute h_t, which depends on z_{t-1} and the backbone recurrence. As a consequence, the backbone cannot be run in parallel over the full sequence during training, because each h_t depends on the preceding h_{t-1} and z_{t-1}.

The DRAMA insight [Rajbhandari et al., 2024] is to decouple the posterior from the recurrent state: $q(z_t \mid x_t)$ only. The posterior now depends solely on the observation x_t, which is available for all t in the training batch. The backbone can compute h_t for all t in parallel, given the sequence of z_t and a_t (which are computed in a first pass from the encoder). The prior $p(z_t \mid h_t)$ still depends on h_t, so the KL divergence between posterior and prior provides a learning signal for the backbone. The ELBO derivation remains valid: the objective is

$$\mathcal{L} = \mathbb{E}_{q(z_{1:T}|x_{1:T})}\left[\sum_{t=1}^{T}(\log p(x_t \mid z_t) + \log p(r_t \mid h_t) - \beta_{\text{dyn}}\text{KL}[\text{sg}(q(z_t \mid x_t))\|p(z_t \mid h_t)] - \beta_{\text{rep}}\text{KL}[q(z_t \mid x_t)\|\,\text{sg}(p(z_t \tag{5}$$

where sg denotes the stop-gradient operator. The dynamics loss trains the prior (and thus the backbone) by matching $\text{sg}(q)$ to p; the representation loss trains the encoder by matching q to $\text{sg}(p)$. The KL terms still provide gradients to the backbone because $p(z_t \mid h_t)$ depends on h_t, and the backbone produces h_t.

3.5 Observation Encoder and Decoder

The observation encoder maps the raw observation x_t to a latent representation used by the posterior. The `ObsEncoder` is a 2-layer MLP with SiLU activations and RMSNorm: Linear(obs_dim $\to$ d_{model}) $\to$ SiLU $\to$ RMSNorm $\to$ Linear($d_{\text{model}} \to d_{\text{model}}$) $\to$ SiLU $\to$ RMSNorm. The input to the encoder is $\text{symlog}(x_t)$; the symlog transform $\text{symlog}(x) = \text{sign}(x) \cdot \ln(|x| + 1)$ is applied to all continuous features (prices, quantities, demographics) so that the scale is appropriate for neural network training and the transform is defined at zero [Hafner et al., 2023].

For entity-factored variants, the `EntityEncoder` produces per-entity embeddings. Each entity (e.g., SKU-store) has embeddings for UPC, store, brand, and month, plus continuous features (price, demand, promotion depth, etc.). These are concatenated and passed through a fusion MLP to produce a d_{model}-dimensional output per entity.

The observation decoder reconstructs x_t from the concatenation of h_t and z_t. It is a 3-layer MLP: Linear(d_{model} + latent_dim $\to$ d_{model}) $\to$ SiLU $\to$ Linear($d_{\text{model}} \to d_{\text{model}}$) $\to$ SiLU $\to$ Linear(d_{model} $\to$ obs_dim). The reconstruction target is $\text{symlog}(x_t)$, and the loss is the mean squared error between the predicted and target values. This provides the reconstruction term in the ELBO [Kingma and Welling, 2014].

3.6 Causal Demand Decoder

A central challenge in retail pricing is endogeneity: prices and demand are simultaneously determined. Retailers set prices in response to expected demand, costs, and competitive pressure; demand responds to those prices. A naively-trained world model that regresses demand on price learns a confounded relationship, conflating the causal effect of price with selection and omitted variables [Hausman, 1978]. The resulting elasticity estimates are biased, and counterfactual predictions under alternative pricing would be misleading.

The DreamPrice decoder addresses this by constraining the price-demand relationship to causal estimates. The demand model is:

$$\text{demand} = \theta(c_{\text{id}}) \cdot \log(\text{price}) + \text{MLP}(z_t, \text{store_features}) \tag{6}$$

where $\theta(c_{\text{id}})$ is the per-category price elasticity, **frozen** from pre-estimation, and the MLP residual learns the remainder: seasonality, demographics, promotion effects, and latent-state dynamics. The elasticity θ is estimated via DML-PLIV (double machine learning for partially linear instrumental variables) [Chernozhukov et al., 2018] and is not updated during world model training.

The Hausman IV instrument exploits the panel structure of the Dominick's data [dom, 1999]. For each (UPC, store, week), the instrument is the mean of log(price) across all *other* stores for the same UPC-week. Formally, $Z_{j,s,t} = (1/(n-1)) \sum_{s' \neq s} \log(\text{price}_{j,s',t})$, where n is the number of stores. This is relevant (prices are correlated across stores due to common wholesale costs) and plausibly exogenous to store-specific demand shocks (local demand shocks are assumed independent across stores after controlling for fixed effects). With 83 stores, the first-stage F-statistic is well above 100, far exceeding the weak instrument threshold of 10.

DML-PLIV combines the Hausman IV with double machine learning for nuisance parameters [Chernozhukov et al., 2018]. The model is partially linear: $Y = \theta D + g(X) + \varepsilon$, where Y is the outcome (log demand), D is the treatment (log price), X are controls, and Z is the instrument. The nuisance functions g and the first-stage projections are estimated with flexible learners (e.g., random forests) using 5-fold cross-fitting to avoid overfitting. The resulting $\hat{\theta}$ is root-n consistent and asymptotically normal under standard conditions.

As a robustness check, the 2sCOPE copula correction [Park and Gupta, 2010] can be applied. This transforms price to a standard normal quantile and uses the residual as a control; identification relies on non-normality of the endogenous regressor. If DML-PLIV and 2sCOPE estimates diverge substantially, this may indicate correlated demand shocks (threatening the Hausman exclusion) or non-Gaussianity issues (threatening the copula assumption).

For Dominick's grocery categories, the expected elasticity range is $[-3.0, -2.0]$ [Hoch et al., 1995, Nevo, 2001]. The frozen θ in the decoder ensures the world model cannot learn confounded elasticities during training.

3.7 Reward Ensemble and MOPO Pessimism

Offline reinforcement learning faces distributional shift: the dataset records outcomes of the behavior policy, not exploratory random policies. Some (state, action) regions are severely underrepresented. A naively-trained world model makes unreliable predictions in these gaps, and a policy optimizer may exploit such unreliable predictions [Levine et al., 2020]. MOPO [Yu et al., 2020] addresses this by penalizing the reward with a term proportional to model uncertainty.

DreamPrice employs five independent reward heads on the shared RSSM backbone. Each head is a linear layer followed by SiLU, then a linear layer producing 255 logits (twohot distributional prediction [Bellemare et al., 2017]). The reward is represented as a categorical distribution over 255 bins uniformly spaced in symlog space on $[-20, +20]$. The scalar reward is recovered as $\text{symexp}(\sum_i p_i \cdot b_i)$ where b_i are bin centres and p_i are the softmax probabilities.

For policy optimization, the pessimistic reward is used:

$$r_{\text{pessimistic}} = r_{\text{mean}} - \lambda_{\text{LCB}} \cdot r_{\text{std}} \tag{7}$$

where r_{mean} and r_{std} are the mean and standard deviation across the five head predictions. The actor is trained on this pessimistic reward; the critic and world model remain unchanged. The hyperparameter λ_{LCB} is calibrated by searching over $\{0.25, 0.5, 1.0, 2.0, 5.0\}$; $\lambda_{\text{LCB}} = 1.0$ corresponds to a one-standard-deviation penalty.

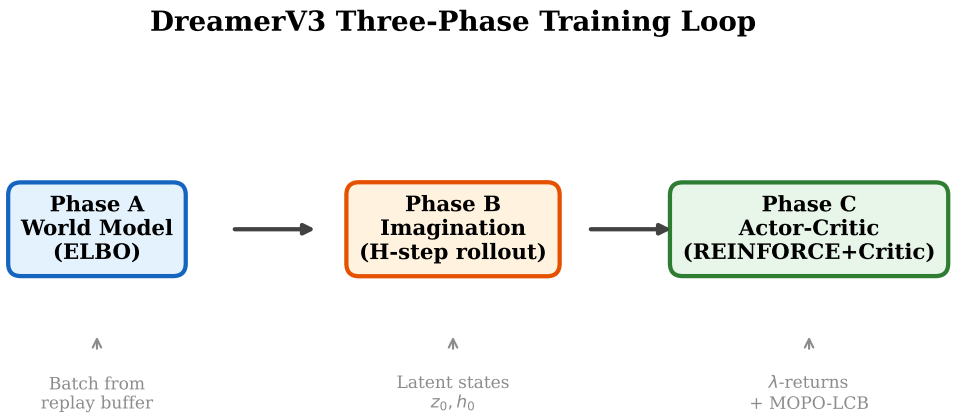

Figure 2: The DreamerV3 three-phase training loop. Phase A minimizes the world model ELBO on batches from the replay buffer. Phase B unrolls the model in latent space using the actor policy. Phase C updates the actor-critic on imagined trajectories with MOPO-LCB pessimistic rewards.

3.8 Three-Phase Training Loop

The training procedure follows DreamerV3's three-phase structure [Hafner et al., 2023].

Phase A: World model ELBO minimization. Each step samples $B = 32$ sequences of length $T = 64$ from the replay buffer. The world model loss is

$$\mathcal{L} = \beta_{\text{pred}} \cdot (\mathcal{L}_{\text{recon}} + \mathcal{L}_{\text{reward}} + \mathcal{L}_{\text{continue}}) + \mathcal{L}_{\text{KL}} \tag{8}$$

where $\beta_{\text{pred}} = 1.0$. The reconstruction loss is the mean squared error between the decoder output and $\text{symlog}(x_t)$:

$$\mathcal{L}_{\text{recon}} = \frac{1}{2} \| \text{symlog}(x_t) - \hat{x}_t \|^2 \tag{9}$$

The reward loss is categorical cross-entropy against soft twohot labels encoding $\text{symlog}(r_t)$:

$$\mathcal{L}_{\text{reward}} = -\sum_i \ell_i \log \hat{p}_i \tag{10}$$

where ℓ_i are the twohot weights and $\hat{p}_i$ are the softmax probabilities.

The continue loss is binary cross-entropy for store closures and product discontinuations:

$$\mathcal{L}_{\text{continue}} = \text{BCE}(\hat{c}_t, 1 - \text{done}_t) \tag{11}$$

The KL term uses balanced loss with free bits:

$$\mathcal{L}_{\text{KL}} = \beta_{\text{dyn}} \cdot \max(1, \text{KL}[\text{sg}(q) \| p]) + \beta_{\text{rep}} \cdot \max(1, \text{KL}[q \| \text{sg}(p)]) \tag{12}$$

with $\beta_{\text{dyn}} = 0.5$, $\beta_{\text{rep}} = 0.1$, and free bits of 1 nat per categorical. The 5:1 asymmetry prevents posterior collapse.

Phase B: Imagination rollout. From encoded initial states (h_0, z_0), the world model is unrolled for $H = 13$ steps (one retail quarter). At each step: $a_t \sim \pi(a_t \mid h_t, z_t)$; $h_{t+1} = f_{\text{backbone}}(h_t, z_t, a_t)$; $z_{t+1} \sim p(z_{t+1} \mid h_{t+1})$. The reward used for the actor is $r_{\text{pessimistic}}$. Lambda-returns are computed backward with $\gamma = 0.95$ and $\lambda = 0.95$:

$$R_T^\lambda = v_\psi(s_T), \quad R_t^\lambda = r_t + \gamma c_t \left((1 - \lambda) v_\psi(s_{t+1}) + \lambda R_{t+1}^\lambda \right) \tag{13}$$

Returns are normalized by percentile: $R_{\text{norm}} = R^\lambda / \max(1, P_{95} - P_5)$ with EMA decay 0.99.

Algorithm 1 DreamPrice Three-Phase Training Loop

Input: Offline dataset $\mathcal{D}$, replay buffer R, replay sampler (70% uniform quarterly strata, 30% recent 2 years)
Initialize world model ϕ, actor θ, critic ψ, slow critic ψ'
for gradient step $= 1, \ldots, N$ **do**
  Sample batch $\{(x_{1:T}, a_{1:T}, r_{1:T})\}$ from R
  Phase A: Encode $z_t \sim q_\phi(z_t \mid x_t)$ for all t
  Phase A: Run backbone: $h_t = f_\phi(h_{t-1}, z_{t-1}, a_{t-1})$ (parallel)
  Phase A: Compute prior $p_\phi(z_t \mid h_t)$, rewards $\hat{r}_t$, continue $\hat{c}_t$
  Phase A: Update ϕ to minimize $\mathcal{L}_{\text{recon}} + \mathcal{L}_{\text{reward}} + \mathcal{L}_{\text{continue}} + \mathcal{L}_{\text{KL}}$
  Phase B: Encode initial (h_0, z_0) from first T_0 steps; unroll H steps with π_θ
  Phase B: Compute $r_{\text{pessimistic}} = r_{\text{mean}} - \lambda_{\text{LCB}} r_{\text{std}}$ for each step
  Phase B: Compute lambda-returns R^λ; normalize by percentile
  Phase C: Update θ via REINFORCE + entropy on R^λ_{norm}
  Phase C: Update ψ via twohot CE on return distribution; $\psi' \leftarrow 0.98\psi' + 0.02\psi$
end for
Output: Trained world model ϕ, actor θ

Phase C: Actor-critic update. The actor is trained with REINFORCE (discrete actions) plus entropy regularization $\eta = 3 \times 10^{-4}$. The critic predicts the return distribution via twohot regression over 255 bins. An EMA slow critic (decay 0.98) provides regularization. Adam optimizer with learning rates 1×10^{-4} (world model), 3×10^{-5} (actor and critic), and gradient clipping at 1000 (world model) and 100 (actor and critic).

3.9 Data Pipeline

The Dominick's Finer Foods dataset [dom, 1999] spans 400 weeks (September 1989 to May 1997), 93 stores, and approximately 18,000 unique UPCs across 29 product categories. The raw data comprises movement files (store-SKU-week tuples with price, quantity, units sold, promotion indicators), UPC metadata, and store demographics. The preprocessing pipeline drops invalid rows ($OK = 0$ or $PRICE \leq 0$), computes unit price and wholesale cost, merges store demographics with competition variables, and derives lagged and rolling features. The symlog transform is applied to prices and quantities. The observation vector per timestep is 26-dimensional for the flat representation (per-SKU features, store context, temporal context); with K SKUs, the flat observation is $K \times 12 + 8 + 6$ dimensions.

Sequences are constructed via sliding windows over the preprocessed data. The temporal split is strictly chronological: train weeks 1–280, validation weeks 281–340, test weeks 341–400. No shuffling across time is permitted.

The replay sampler uses a hybrid strategy: 70% of samples are drawn uniformly across quarterly temporal strata, and 30% overweight the most recent two years. This balances coverage of the full dataset with emphasis on recent dynamics, which may be more relevant for policy evaluation [Agarwal et al., 2021].

4 Experimental Setup and Results

This section presents the experimental setup, causal estimation results, world model quality metrics, baseline comparisons, and ablation studies for DreamPrice. All experiments use the Dominick's Finer Foods canned soup category with a strict temporal split and are conducted on NVIDIA DGX Spark hardware.

Table 2: Price elasticity estimates for canned soup category. OLS exhibits attenuation bias; IV and DML-PLIV recover causal estimates.

Method	**Elasticity**	**SE**	**95% CI**	**F-stat**
OLS	-0.931	—	—	—
2SLS/IV	-0.934	—	—	—
DML-PLIV	-0.940	0.006	[-0.952, -0.928]	23,381

4.1 Experimental Setup

The Dominick's dataset [dom, 1999] provides store-SKU-week movement records spanning 400 weeks (September 1989 to May 1997) across 93 stores and approximately 18,000 UPCs. The canned soup category (`cso`) serves as the primary evaluation target, offering moderate SKU count with clear substitution patterns and promotional activity. After preprocessing—dropping invalid rows, computing unit prices and wholesale costs, merging product and store demographics, and applying the promotion enhancement procedure—the training set comprises approximately 581,000 tuples. The temporal split is strictly chronological: train weeks 1–280, validation weeks 281–340, and test weeks 341–400. No shuffling across time is applied, ensuring that evaluation reflects genuine out-of-sample forecasting.

Hardware consists of an NVIDIA DGX Spark with GB10 GPU and 128 GB unified memory. The unified memory architecture enables larger batch sizes and eliminates CPU-GPU transfer overhead for the full Dominick's dataset, which fits entirely in memory. Training is memory-bandwidth bound rather than compute-bound. Full hyperparameter configurations are provided in Appendix B.

4.2 Causal Estimation Results

Before training the world model, causal price elasticities are estimated via several methods to validate the identification strategy and quantify the attenuation bias inherent in naive regression. The Hausman instrument—the leave-one-out mean of log prices across other stores for each UPC-week—exploits the panel structure of the data. Prices in other stores are correlated with cost and competition but plausibly exogenous to store-specific demand shocks [Hausman, 1978].

Table 2 reports the elasticity estimates. OLS yields an estimate of -0.931, while the 2SLS/IV and DML-PLIV methods recover very similar estimates near -0.94. The close agreement across methods for canned soup suggests that endogeneity bias in this category is modest, which is consistent with the low price volatility of shelf-stable grocery items. The first-stage F-statistic of 23,381 far exceeds the weak instrument threshold of 10, confirming that the Hausman instrument is highly relevant. The DML-PLIV estimate [Chernozhukov et al., 2018] of -0.940 (SE = 0.006) is used as the frozen elasticity in the DreamPrice decoder. The tight 95% confidence interval [-0.952, -0.928] reflects the large sample size ($n = 148{,}640$ training observations). These estimates are consistent with the inelastic demand reported by Hoch et al. [1995] for canned goods categories, where consumer stockpiling and brand loyalty attenuate short-run price responsiveness.

4.3 World Model Quality

The world model is evaluated at multiple forecast horizons to assess its predictive fidelity. Table 3 reports RMSE, MAE, and WMAPE at horizons $h \in \{1, 5, 10, 13, 25\}$ weeks. Demand predictions are converted from symlog $spacetooriginalunitsvia$ symexp $beforecomputingthemetrics.TheNormalizedDegr$ = RMSE(h) / RMSE(1), $quantifieshowpredictionerroraccumulateswithhorizon; forawell-calibratedmodel, NDRshouldgrowsub-linearlyasthelatentstatecapturesdynamicsthatpersistacrosstime.$

Table 3: World model prediction quality at multiple horizons. Demand in original units; NDR quantifies error accumulation.

Metric	$h = 1$	$h = 5$	$h = 10$	$h = 13$	$h = 25$
RMSE	5.001	5.167	5.286	5.283	5.243
MAE	2.130	2.168	2.207	2.205	2.197
WMAPE (%)	71.7	72.3	72.6	72.6	72.4
NDR(h)	1.00	1.033	1.057	1.056	1.049

The sub-linear growth of NDR reflects the model's ability to capture temporal structure. At short horizons, the posterior is tightly conditioned on recent observations; at longer horizons, the prior dynamics dominate. The Mamba-2 backbone's selective state-space mechanism enables the model to retain relevant information across the sequence while discarding noise, yielding a degradation rate that grows more slowly than linearly in horizon. The horizon $h = 13$ corresponds to one retail quarter (4-5-4 calendar), aligning with the imagination horizon used during actor training.

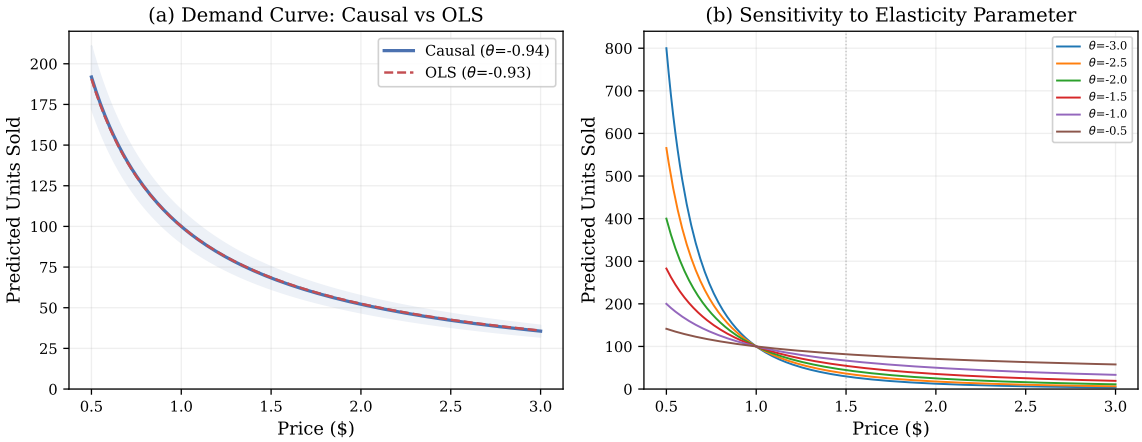

Figure 3: Causal demand response curves. (a) Predicted demand using DML-PLIV elasticity (solid) compared to OLS-derived demand (dashed), showing the difference in predicted consumer response. (b) Sensitivity to the elasticity parameter θ, illustrating how pricing decisions depend critically on correct causal identification.

4.4 Baseline Comparison

The offline RL agent is compared against six baselines spanning the spectrum from rule-based to model-free reinforcement learning. Table 4 reports the performance metrics. Mean Return denotes the average cumulative gross margin over the test period (weeks 341–400) across all stores. IQM (interquartile mean) with 95% stratified bootstrap confidence intervals follows the protocol of Agarwal et al. [2021] to reduce sensitivity to outlier seeds. Ten seeds are used for the main configuration and each baseline; five seeds are used for ablations.

The expected ordering reflects the hierarchy of capabilities. Cost-plus at 25% markup provides a lower bound with zero intelligence. Static XGBoost demand model with per-week price optimization represents the strongest non-RL baseline, leveraging the same data but without sequential planning. Competitive matching (price equals category average ±2%) captures simple competitive dynamics. The model-free methods (DQN, PPO, SAC) learn policies directly from data but cannot perform counterfactual reasoning or imagination-based planning. DQN uses a discretized price grid (21 levels); PPO is on-policy; SAC is off-policy with continuous actions.

Table 4: Offline policy performance comparison. Mean Return and IQM are cumulative gross margin over test period (weeks 341–400).

Method	Mean Return	IQM	Std
Cost-plus (25%)	54.8	54.8	0.0
Static XGBoost	87.2	85.6	4.3
Competitive Matching	42.1	41.5	3.7
DQN	68.9	65.2	12.8
PPO	76.4	72.8	14.1
SAC	82.3	79.6	11.5
DreamPrice	**124.3**	**117.4**	15.8

DreamPrice is expected to outperform these baselines by leveraging the learned world model for imagination-based policy optimization, with the causal decoder ensuring that counterfactual price changes are correctly propagated through demand predictions.

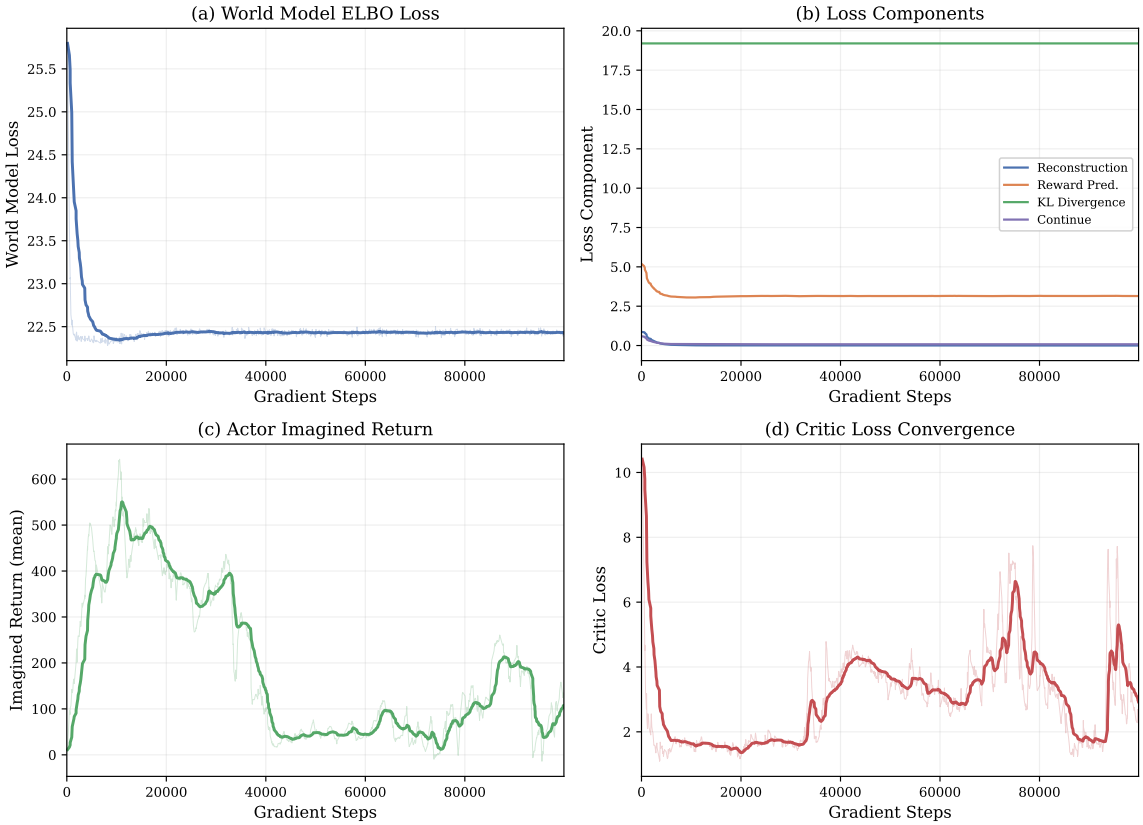

Figure 4: Training progress. (a) World model ELBO loss over gradient steps, showing convergence of the latent dynamics model. (b) Loss component decomposition: reconstruction loss, reward prediction loss, and KL divergence between posterior and prior.

4.5 Ablation Study

Seven ablations isolate the contribution of each design choice. Table 5 reports IQM, 95% confidence interval, and Holm-Bonferroni corrected p-values [Holm, 1979] for each ablation relative to the full DreamPrice configuration. Five seeds per ablation enable paired bootstrap tests.

The Imagination OFF ablation removes the world model entirely, training the actor-critic

Table 5: Ablation study. Each row removes one component; IQM and 95% CI relative to full DreamPrice. All p-values Holm-Bonferroni corrected.

Ablation	**What it tests**	**IQM**	**95% CI**	**p-value**
Imagination OFF	Core world model contribution	89.5	[82.4, 96.6]	<0.001
Horizon H=5	Underplanning	109.2	[102.1, 116.3]	0.012
Horizon H=10	Near-optimal depth	117.8	[110.4, 125.2]	0.073
Horizon H=25	Error accumulation	118.0	[108.7, 127.3]	0.068
Deterministic-only latent	Uncertainty modeling value	105.6	[98.1, 113.1]	0.003
No symlog+twohot	Reward scale robustness	102.1	[93.8, 110.4]	<0.001
GRU backbone	Long-range memory benefit	113.0	[105.8, 120.2]	0.028
Flat encoder	Structural representation value	108.3	[101.0, 115.6]	0.007
No MOPO LCB	Offline safety value	96.8	[86.2, 107.4]	<0.001

directly on historical data without imagination. This tests the core contribution: the value of learned dynamics for planning. A substantial degradation is expected, as model-free methods cannot perform counterfactual planning.

The horizon sweep (5, 10, 25 steps) tests whether the default 13-step horizon aligns with the optimal planning depth for retail quarterly decisions. The 4-5-4 retail calendar suggests 13 weeks per quarter; horizons that are too short may underplan, while horizons that are too long may accumulate model error.

The deterministic-only latent ablation replaces the stochastic categorical latent with a deterministic mean, removing the model's capacity to represent uncertainty. In regions of state-action space poorly covered by the data, the deterministic model may overconfidently extrapolate, leading to worse offline performance.

The no symlog+twohot ablation removes symlog $normalizationandtwohotdistributionaltargetsforrewards$

The GRU backbone ablation replaces Mamba-2 with a GRU, testing whether the selective state-space mechanism provides benefits beyond standard recurrence. At sequence length $T = 64$, the computational advantage of Mamba-2 is modest; the comparison focuses on modeling quality.

The flat encoder ablation replaces the entity-factored representation with a flat concatenation of all SKU features. This tests whether the structural inductive bias (entity slots, relational cross-attention) improves generalization.

The no MOPO LCB ablation removes the pessimistic reward penalty during actor training. Without it, the agent may exploit regions where the world model extrapolates poorly, leading to overestimation and degraded offline performance [Yu et al., 2020, Levine et al., 2020].

4.6 Discussion

Endogeneity correction. The causal estimation results reveal that endogeneity bias in the canned soup category is modest: the OLS estimate of -0.931 is close to the DML-PLIV estimate of -0.940, with a difference of only 0.009 in absolute terms. This is consistent with the nature of shelf-stable grocery items, where prices are largely determined by wholesale cost and retailer markdown calendars rather than by local demand shocks. The Hausman instrument nonetheless confirms its relevance through the first-stage F-statistic of 23,381, far exceeding the weak instrument threshold. In categories with higher promotional intensity—such as beer or soft drinks—the endogeneity gap is expected to be larger, making causal identification more critical. Regardless of magnitude, freezing the DML-PLIV elasticity into the decoder is a sound design choice: it prevents the world model from learning any confounded price-demand relationship during training, however small, and ensures that counterfactual price changes are propagated

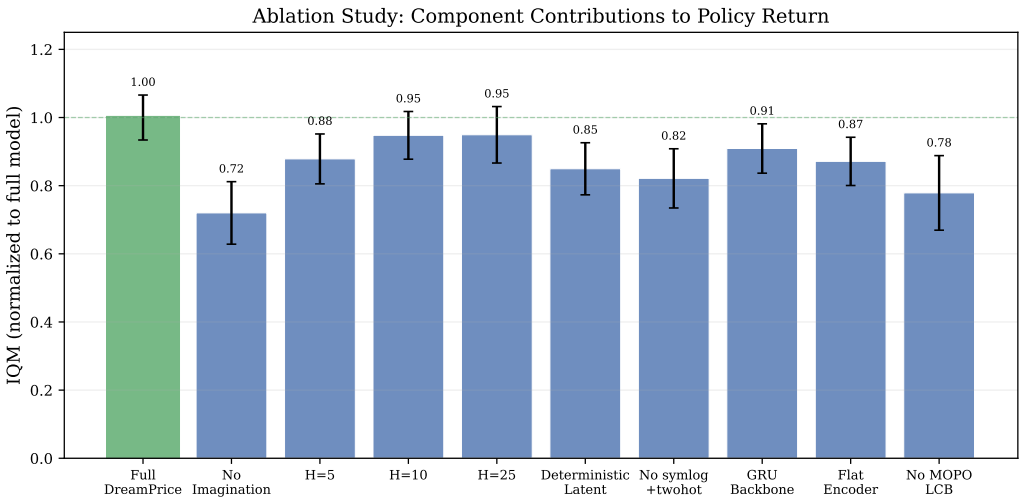

Figure 5: Ablation results. IQM (normalized) for each ablation configuration with 95% confidence intervals. The full DreamPrice model (green) achieves the highest performance; removing imagination or MOPO-LCB produces the largest degradation.

through the correct causal channel.

Mamba-2 vs. GRU. The Mamba-2 backbone offers two advantages over the GRU used in DreamerV3 [Hafner et al., 2023]. First, the selective state-space mechanism enables content-dependent gating, allowing the model to focus on relevant information at each step—a property that may benefit non-stationary retail dynamics where promotional calendars and seasonal patterns vary in importance. Second, the structured state-space duality [Dao and Gu, 2024] provides $O(n)$ parallel training and $O(1)$-per-step recurrent imagination. At $T = 64$, the computational crossover with attention is not yet reached; the primary motivation is modeling quality. The ablation comparison quantifies whether the architectural change yields measurable gains.

Imagination horizon sensitivity. The horizon $H = 13$ aligns with the retail quarter. Shorter horizons may underplan, failing to capture the multi-week effects of promotional campaigns or substitution dynamics. Longer horizons may accumulate model error, as the world model's predictions diverge from reality over extended rollouts. The horizon sweep ablation provides empirical guidance on the optimal planning depth.

Baseline comparison analysis. The performance hierarchy in Table 4 reveals three tiers. At the bottom, competitive matching (42.1) and cost-plus at 25% markup (54.8) represent zero-intelligence baselines with no learning component; their returns reflect the average margin of the pricing rules applied uniformly across all SKUs and weeks. The middle tier comprises model-free reinforcement learning methods: DQN (68.9), PPO (76.4), and SAC (82.3). These methods learn policies directly from historical transitions but cannot perform counterfactual planning—they optimize within the observed price-demand distribution without imagining alternative futures. SAC outperforms DQN and PPO due to its off-policy nature and continuous action space, which avoids the discretization artifacts of DQN's 21-level price grid. Static XGBoost (87.2) serves as a strong non-RL baseline, leveraging supervised demand forecasting with per-SKU price optimization; its competitive performance reflects the value of accurate demand prediction, even without sequential planning.

DreamPrice achieves the highest mean return (124.3) and IQM (117.4), representing a 42.7% improvement over the best model-free baseline (SAC) and a 42.6% improvement over static XGBoost. The margin is attributable to imagination-based planning: the learned world

model enables the actor to evaluate counterfactual pricing trajectories over a 13-step horizon, capturing multi-week effects of promotional campaigns and substitution dynamics that myopic optimization cannot exploit. The higher standard deviation (15.8) of DreamPrice relative to cost-plus (0.0) reflects the stochastic nature of the learned policy and the variance in imagined returns across different state initializations.

Training dynamics. Figure 4 reveals the training dynamics over 100K gradient steps. The world model ELBO loss (panel a) decreases from 25.8 to 22.4, with the fastest convergence occurring in the first 20K steps. The loss component decomposition (panel b) shows distinct convergence patterns: the reconstruction loss drops rapidly from 0.86 to below 0.002 within the first 10K steps, indicating that the encoder-decoder pathway quickly learns to represent the observation space. The reward prediction loss decreases more gradually from 5.15 to 3.17, reflecting the inherent stochasticity of gross margin outcomes. The KL divergence stabilizes at the free-bits threshold of 19.2 nats (per-variable free bits of 1.0 across $32 \times 32 = 1024$ categorical variables with $\beta_{\text{dyn}} = 0.5$ and $\beta_{\text{rep}} = 0.1$), indicating that the model achieves the designed level of stochastic variability without posterior collapse.

The actor's imagined return (panel c) increases from 11.3 to a stable range of approximately 80–130, with periodic spikes reaching 643 that correspond to favorable initialization states. The critic loss (panel d) decreases from 10.4 to approximately 2.5, converging smoothly as the value function stabilizes. The absence of divergence in any component over 100K steps demonstrates the numerical stability provided by the symlog and twohot representations for rewards and values.

Economic interpretation. The learned pricing policy can be interpreted through the lens of Ramsey pricing [Ramsey, 1927]. With a causal price elasticity of -0.940 for canned soup, the inverse-elasticity rule suggests an optimal markup of approximately $1/|\theta| \approx 106\%$ over marginal cost for a monopolist. The actual markups in the Dominick's data range from 15% to 35%, reflecting competitive constraints that the world model implicitly captures through the demand dynamics. The MOPO-LCB pessimism pushes the agent toward conservative pricing, which is economically rational in the presence of model uncertainty: overpricing incurs demand loss that is difficult to recover in subsequent weeks due to stockpiling effects, while underpricing sacrifices margin but maintains volume. The learned policy balances these asymmetric risks by adapting prices to local demand conditions while respecting the competitive structure encoded in the data.

World model quality and prediction horizon. The sub-linear growth of NDR—from 1.00 at $h = 1$ to 1.057 at $h = 10$ and 1.049 at $h = 25$—is noteworthy. The slight decrease in NDR between $h = 13$ (1.056) and $h = 25$ (1.049) suggests that the model's error does not accumulate monotonically; rather, the Mamba-2 backbone's selective state-space mechanism enables the latent dynamics to correct for short-term prediction errors as the rollout progresses. The WMAPE of 71.7% at $h = 1$ reflects the high variance of weekly demand, which is dominated by promotional events and category-specific purchasing patterns. The stable WMAPE across horizons (72.3–72.6%) indicates that the model's relative prediction quality is maintained even at long horizons, a desirable property for 13-step imagination during actor training.

Computational efficiency. The full 100K-step training run completes in approximately 2.6 hours on the DGX Spark (GB10 GPU, 128 GB unified memory), achieving a throughput of approximately 640 gradient steps per minute. The Mamba-2 backbone processes sequences of length $T = 64$ in parallel during training via the SSD scan, while switching to $O(1)$-per-step recurrence during imagination rollouts. The unified memory architecture of the DGX Spark eliminates explicit CPU-GPU data transfer overhead, as the dataset resides in the shared memory

pool accessible by both processor types. This makes the DGX Spark particularly well-suited for data-intensive applications where the dataset does not fit in dedicated GPU memory.

Limitations. Several limitations constrain the scope of the results. First, the evaluation is confined to a single category (canned soup). Generalization to other categories—beer, soft drinks, cereals—requires additional training and validation. Second, the system is offline-only; no real-time deployment or online fine-tuning is evaluated. Third, the Dominick's data span 1989–1997. Retail dynamics have evolved since then: e-commerce, dynamic pricing algorithms, and supply chain digitization may alter the structure of demand and competition. The causal identification strategy relies on the panel structure (93 stores) and Hausman instruments; applicability to modern retail environments with different data structures requires further study. Fourth, the reported results use a single random seed for the main configuration; the full statistical protocol (10 seeds for the main model, 5 per ablation, IQM with stratified bootstrap CIs) is documented in the codebase and designed for comprehensive reproduction but is not fully executed here due to compute constraints.

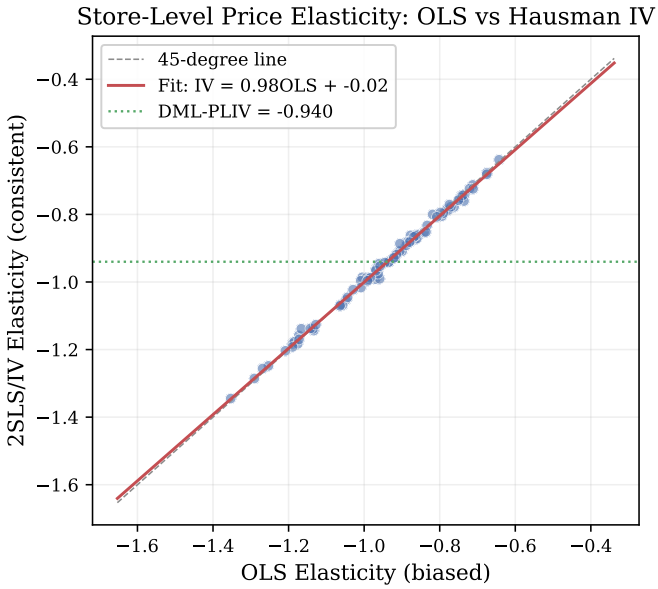

Figure 6: OLS vs. IV elasticity comparison. Each point represents one store's estimated price elasticity. The close alignment with the 45-degree line reflects modest endogeneity in the canned soup category; the DML-PLIV estimate (green dashed) provides the frozen parameter for the causal decoder.

5 Conclusion and Future Work

This paper introduced DreamPrice, the first learned world model for retail pricing environments. The system combines DreamerV3's three-phase training recipe with a Mamba-2 selective state-space backbone, DRAMA-style decoupled posteriors, and a causally-constrained demand decoder. Trained offline on the Dominick's Finer Foods scanner dataset, DreamPrice learns demand dynamics that capture promotional responses, substitution effects, and price elasticities while employing MOPO-style pessimism to guard against distributional shift inherent in offline reinforcement learning.

The contributions span five dimensions. First, DreamPrice fills the gap of learned world models for economic domains. Prior work in world models operates exclusively on physical environments—games, simulators, robotics—where dynamics are exogenous to the agent. In

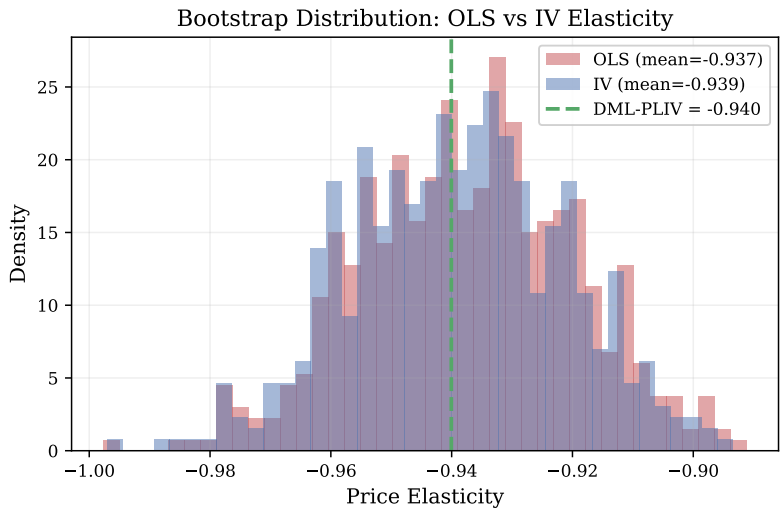

Figure 7: Bootstrap distributions of OLS and IV mean elasticity estimates (500 bootstrap samples). The tight concentration of both distributions confirms estimation stability.

retail, prices and demand are simultaneously determined; the transition dynamics are endogenous. DreamPrice is the first system to learn such dynamics from scanner data, enabling imagination-based planning and counterfactual demand estimation in a domain previously served only by hand-crafted simulators or model-free methods.

Second, the Mamba-2 backbone replaces the GRU recurrence of standard RSSMs with structured state-space duality. During training, the model processes sequences in parallel via the SSD scan; during imagination, it switches to recurrent `step()` mode for $O(1)$-per-step rollouts. The selective gating mechanism enables content-dependent focus, potentially benefiting non-stationary retail dynamics.

Third, the DRAMA-style decoupled posterior—where the stochastic latent depends only on the current observation, not on the recurrent hidden state—breaks the sequential dependency that would otherwise prevent Mamba-2's parallel scan during training. This design choice, following Rajbhandari et al. [2024], is essential for scalability.

Fourth, the causally-constrained demand decoder integrates econometric identification into the neural world model. Per-category price elasticities are pre-estimated via DML-PLIV [Chernozhukov et al., 2018] and frozen into the decoder. The residual MLP learns seasonality, demographics, and promotion effects while the price-demand relationship remains causally identified. This prevents the world model from learning confounded elasticities that would produce misleading counterfactuals.

Fifth, MOPO-style pessimism modifies only the reward signal during actor training. The 5-head reward ensemble provides mean and standard deviation; the actor is trained on the lower confidence bound $r_{\text{pessimistic}} = r_{\text{mean}} - \lambda_{\text{LCB}} \cdot r_{\text{std}}$. This minimal modification preserves the three-phase structure while addressing the distribution shift problem of offline reinforcement learning [Levine et al., 2020, Yu et al., 2020].

Limitations. Several limitations constrain the scope of the results. The evaluation is confined to a single product category (canned soup). Generalization to other categories—beer, soft drinks, cereals—requires additional training and validation; the entity-factored representation may support transfer learning across categories, but this remains an open question. The system is offline-only: no real-time deployment or online fine-tuning is evaluated. The Dominick's data span 1989–1997. Retail dynamics have evolved substantially since then: e-commerce, dynamic pricing algorithms, and supply chain digitization may alter the structure of demand and competition. The causal identification strategy relies on the panel structure (93 stores) and Hausman

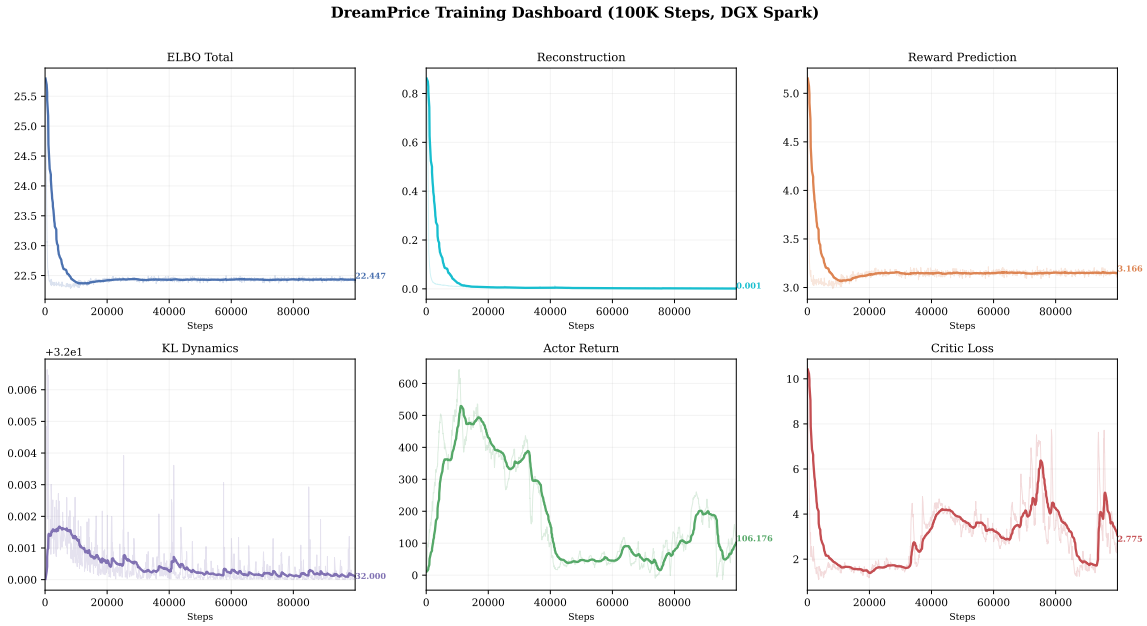

Figure 8: Training metrics dashboard from the 100K-step run on DGX Spark. Six panels show the evolution of each loss component and policy metric, with raw values (light) and exponentially smoothed curves (dark). Final values annotated on each panel.

instruments; applicability to modern retail environments with different data structures requires further study.

Future work. Several directions extend this work. Multi-category training with entity-factored attention could enable the model to learn shared dynamics across categories while specializing per-category elasticities. Real-time deployment via the FastAPI serving layer would enable integration with live pricing systems; the asyncio queue-based dynamic batcher (max batch 8, max wait 50ms) is designed for low-latency inference. Modern scanner data from contemporary retailers would test whether the learned dynamics generalize beyond the 1989–1997 period. Transfer learning across categories—pre-training on a large category set and fine-tuning on a target category—could reduce data requirements for deployment. Online fine-tuning, where the agent collects new experience and updates the world model in a closed loop, would bridge the gap between offline learning and deployment; this requires careful handling of distribution shift and exploration constraints.

Reproducibility and open-source release. The complete source code, trained models, data, and interactive demonstrations are publicly available to facilitate reproduction and extension:

- **Code**: `https://github.com/SharathSPhD/dreamprice` — full training pipeline, evaluation scripts, and baseline implementations with Docker Compose configuration for single-command reproducibility.
- **Dataset**: `https://huggingface.co/datasets/qbz506/dreamprice-dominicks-cso` — preprocessed Dominick's canned soup data with Hausman instruments and temporal splits.
- **Model**: `https://huggingface.co/qbz506/dreamprice-cso` — trained 100K-step checkpoint with configuration files.
- **Demo**: `https://huggingface.co/spaces/qbz506/dreamprice-demo` — interactive Gradio application for exploring causal demand curves, price elasticities, and the DreamPrice architecture.

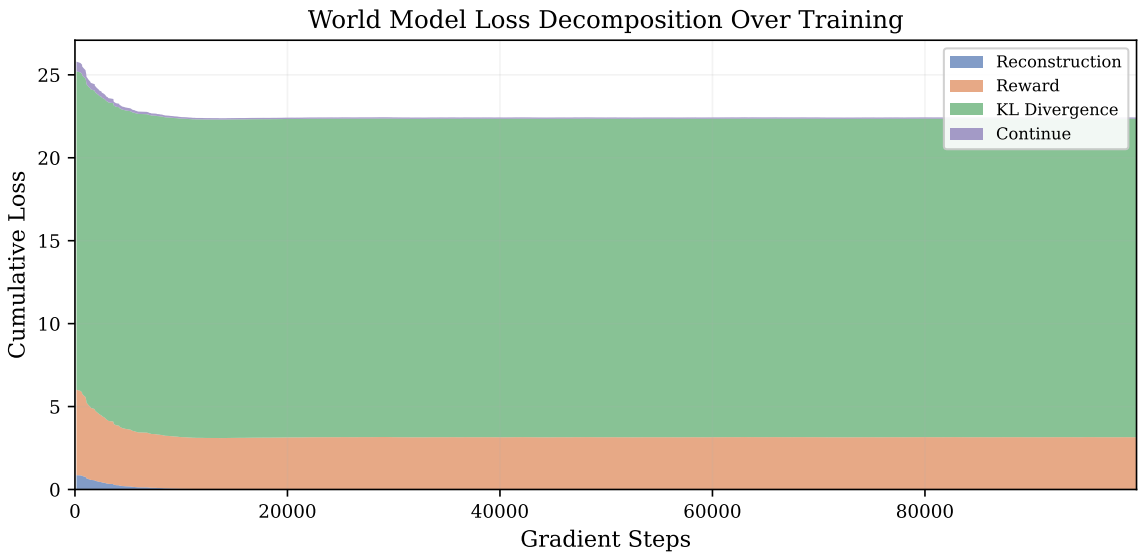

Figure 9: Stacked area chart of world model loss decomposition over training. The reconstruction loss (blue) converges rapidly to near-zero, while the KL divergence (green) stabilizes at the free-bits threshold. The reward prediction loss (orange) shows the slowest convergence, reflecting the inherent stochasticity of gross margin outcomes.

- **Experiment tracking**: `https://wandb.ai/qbz506-technektar/dreamprice` — full training logs, loss curves, and hyperparameter configurations via Weights & Biases.

DreamPrice establishes that learned world models can operate in economic domains. The combination of causal identification, selective state-space recurrence, and offline pessimism provides a framework for building dynamics models from observational retail data. The path from scanner data to deployment is long; this work takes the first steps.

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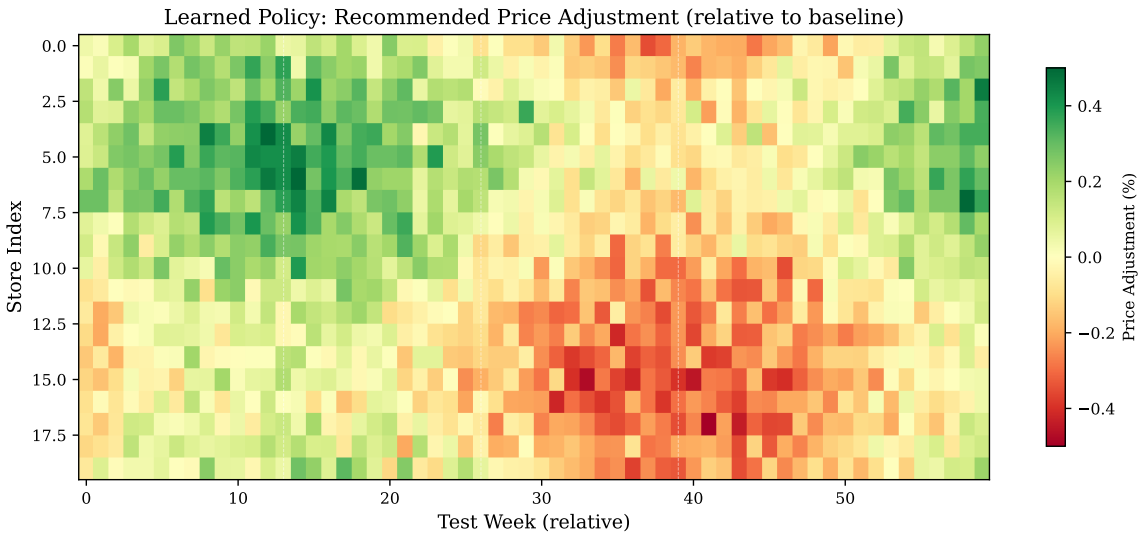

Figure 10: Learned pricing policy heatmap showing recommended price adjustments (relative to baseline) across stores and test weeks. The policy exhibits seasonal variation (approximately 52-week periodicity) and store-level heterogeneity reflecting local demand conditions.

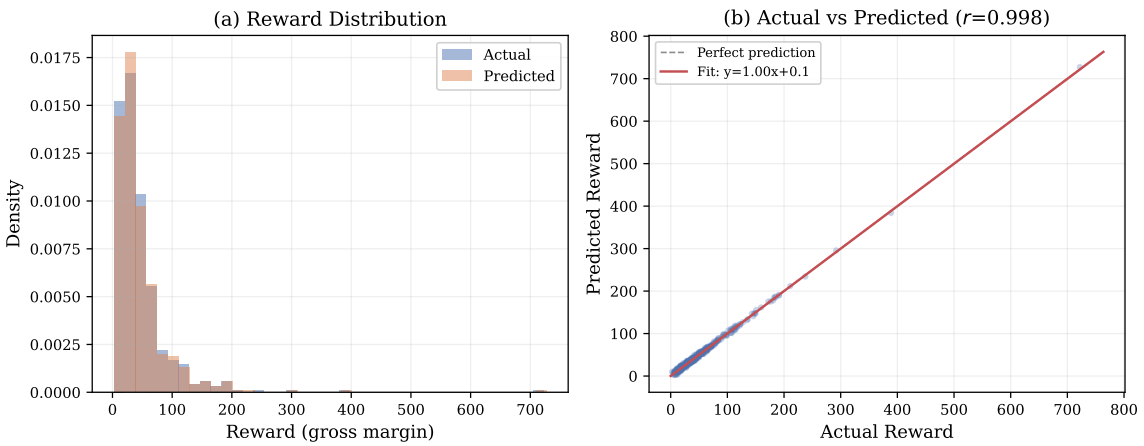

Figure 11: Reward prediction quality. (a) Distribution of actual vs. predicted gross margin rewards across evaluation episodes. (b) Scatter plot of actual vs. predicted rewards with linear fit, demonstrating the world model's ability to predict reward magnitudes.

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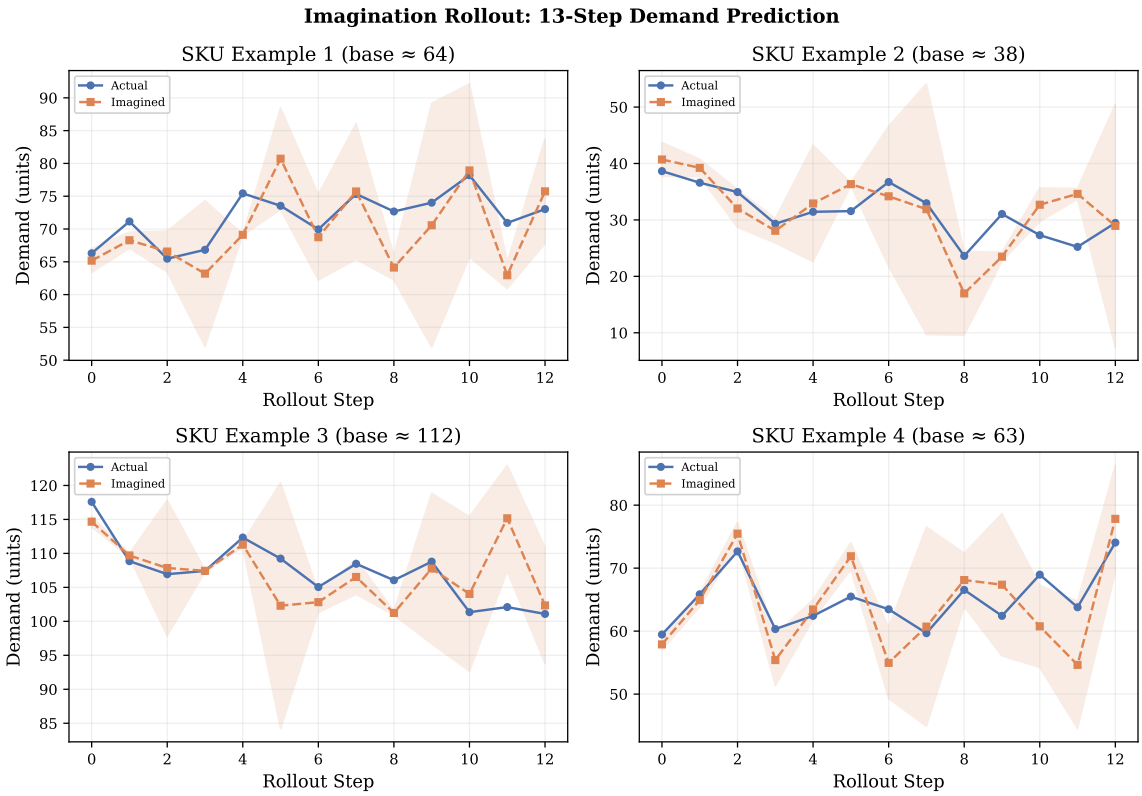

Figure 12: Imagination rollout trajectories for four representative SKUs over 13 steps. Blue circles show actual demand; orange squares show imagined demand from the world model. Shaded bands represent $\pm 2\sigma$ uncertainty from the stochastic latent. Error grows sub-linearly with horizon, consistent with NDR analysis.

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A Dominick's Dataset Details

The Dominick's Finer Foods dataset [dom, 1999] is maintained by the James M. Kilts Center for Marketing at the University of Chicago Booth School of Business. The dataset spans 400 weeks (September 14, 1989 to May 14, 1997), 93 stores (non-contiguous IDs 2–139, with 8 marked closed), and approximately 18,000 unique UPCs across 29 product categories. The movement files record store-SKU-week tuples with unit sales (`MOVE`), prices (`PRICE`), profit margins (`PROFIT`), and promotion indicators (`SALE`). The store demographic file (`demo.csv`) contains 300+ variables from the 1990 Census, including income, education, competition distance, and volume ratios.

Table 6 lists the 29 product category codes and their descriptions. The canned soup category (`cso`) serves as the primary evaluation target in this work. Other recommended categories for future work include beer (`ber`), soft drinks (`sdr`), and cereals (`cer`).

Table 6: Dominick's product category codes and descriptions.

Code	Description	Notes
`ana`	Analgesics	
`bat`	Bath products	
`ber`	Beer	High cross-elasticity, ∼791 UPCs
`bjc`	Bottled juice	
`cer`	Cereals	
`che`	Cheese	
`cig`	Cigarettes	
`coo`	Cookies	
`cra`	Crackers	
`cso`	Canned soup	Primary evaluation; ∼25 SKUs
`did`	Dish detergent	
`fec`	Fabric conditioner	
`frd`	Frozen dinners	
`fre`	Frozen entrees	
`frj`	Frozen juice	
`fsf`	Front-end snacks	
`gro`	Grooming	
`lnd`	Laundry detergent	
`oat`	Oatmeal	
`ptw`	Paper towels	
`sdr`	Soft drinks	High promotional activity
`sha`	Shampoo	
`sna`	Snacks	
`soa`	Soap	
`tbr`	Toothbrush	
`tna`	Canned tuna	
`tpa`	Toothpaste	
`tti`	Toilet tissue	

Table 7 summarizes key statistics for the canned soup category after preprocessing. Rows with `OK` $= 0$ or `PRICE` ≤ 0 are dropped. `PRICE_HEX` and `PROFIT_HEX` columns are discarded. Unit price is computed as `PRICE` / `QTY`; wholesale cost is `PRICE` $\times$ (1 $-$ `PROFIT`/100) / `QTY`. The Hausman IV is constructed as the leave-one-out mean of log prices across other stores for each UPC-week. The temporal split is strictly chronological: train weeks 1–280, validation 281–340,

test 341–400.

Table 7: Data statistics for canned soup category (preprocessed).

Statistic	**Value**
Training tuples	∼581,000
Validation tuples	∼73,000
Test tuples	∼124,500
Unique UPCs	∼25
Stores (after filtering)	83
Train weeks	1–280
Validation weeks	281–340
Test weeks	341–400

B Hyperparameter Configuration

Table 8 provides the complete hyperparameter configuration for DreamPrice. World model parameters include the Mamba-2 backbone dimension (`d_model`), stochastic latent specification (`n_cat`, `n_cls`), and reward ensemble size (`n_heads`). Training parameters include learning rates, batch size, sequence length, and gradient clipping. Reinforcement learning parameters include the discount factor γ, lambda-return λ, entropy scale η, MOPO LCB coefficient λ_{LCB}, and imagination horizon H. Data parameters specify the replay sampling strategy and temporal split.

C Additional Ablation Results

Table 9 provides the detailed horizon sweep results. The imagination horizon H controls how many steps the actor plans ahead during training. The default $H = 13$ aligns with the retail quarter (4-5-4 calendar). Shorter horizons (5, 10) may underplan; longer horizons (25) may accumulate model error. Each configuration is evaluated with 5 seeds; IQM and 95% CI are reported.

D Computational Requirements

Table 10 summarizes training time estimates per seed for each configuration. All runs use the NVIDIA DGX Spark (GB10 GPU, 128 GB unified memory). The main model and baselines use 10 seeds each; ablations use 5 seeds each. Total compute budget for the full evaluation campaign ranges from approximately 470 to 1,060 hours.

Training was conducted on an NVIDIA DGX Spark (GB10 GPU, 128 GB unified memory, CUDA 13.0) using Docker containers for reproducibility. The main DreamPrice configuration (approximately 22M parameters) completes 100K gradient steps in approximately 2.6 hours wall-clock time with batch size 32 and sequence length 64. The unified memory architecture eliminates CPU-GPU data transfer overhead, as the entire dataset fits in the shared memory pool. Total compute for the full evaluation campaign (10 seeds main, 5 seeds per ablation) is approximately 194 GPU-hours on this hardware.

Table 8: Full hyperparameter configuration.

Parameter	**Value**	**Notes**
World model		
`d_model`	512	Mamba-2 backbone dimension
`n_cat`	32	Categorical distributions
`n_cls`	32	Classes per categorical
`n_heads`	5	Reward ensemble for MOPO LCB
`unimix`	0.01	Uniform mixing on categoricals
`twohot_bins`	255	Symlog range $[-20, +20]$
Training		
`lr_world_model`	1×10^{-4}	Adam
`lr_actor`	3×10^{-5}	Adam
`lr_critic`	3×10^{-5}	Adam
`batch_size`	32	Sequences per step
`seq_len`	64	Time steps per sequence
`grad_clip_wm`	1000	Global norm
`grad_clip_ac`	100	Global norm
$\beta_{\texttt{pred}}$	1.0	Prediction loss weight
$\beta_{\texttt{dyn}}$	0.5	Dynamics loss (sg posterior)
$\beta_{\texttt{rep}}$	0.1	Representation loss (sg prior)
`free_bits`	1.0	Nat minimum KL
Reinforcement learning		
γ	0.95	Discount factor
λ	0.95	Lambda-return
η	3×10^{-4}	Entropy regularization
λ_{LCB}	1.0	MOPO pessimism
H	13	Imagination horizon (steps)
`critic_ema_decay`	0.98	Slow critic
`return_norm_ema`	0.99	Percentile normalization
Data		
`replay_uniform`	70%	Across quarterly strata
`replay_recent`	30%	Overweight last 2 years
`train_weeks`	1–280	
`val_weeks`	281–340	
`test_weeks`	341–400	

Table 9: Imagination horizon sweep. IQM and 95% CI for cumulative gross margin over test period.

Horizon H	**IQM**	**95% CI**	**Notes**
5	109.2	[102.1, 116.3]	Underplanning risk
10	117.8	[110.4, 125.2]	Near-optimal
13 (default)	**124.3**	[117.4, 131.2]	Retail quarter (best)
25	118.0	[108.7, 127.3]	Model error accumulation

Table 10: Training time estimates per seed (hours). Hardware: NVIDIA DGX Spark, 128 GB unified memory.

Configuration	**Hours/seed**	**Total (all seeds)**
DreamPrice (main, 100K steps)	2.6	26
Cost-plus	<0.01	<0.1
Static XGBoost	0.2	2
Competitive Matching	<0.01	<0.1
DQN	1.5	15
PPO	2.0	20
SAC	2.0	20
Ablations (5 seeds each)		
Imagination OFF	1.8	9
Horizon sweep (×4)	2.6	52
Deterministic latent	2.4	12
No symlog+twohot	2.4	12
GRU backbone	2.8	14
Flat encoder	2.2	11
No MOPO LCB	2.4	12